Pranay Katari

469-571-2396 | pranaykatari001@gmail.com | linkedin.com/in/pranaykatari | github.com/Pranay-Katari | Portfolio

Education

The University of Texas at Austin — Austin, TX
Bachelor of Science in Computer Science — *May 2028*

- Coursework: Data Structures, Discrete Mathematics, Computer Architecture, Linear Algebra, Statistics, Operating Systems, Algorithms, Machine Learning

Experience

Software Engineer Intern — June 2026 – August 2026
Invesco Ltd. — *Houston, TX*

- Automated **AWS EC2** gateway provisioning with **Terraform** and **Bitbucket Pipelines**, supporting data access for **4,800+ employees** across EMEA.
- Integrated **AWS Secrets Manager** and contributed **Python Lambda** code to a ServiceNow-to-Snowflake workflow, cutting request processing from **1–2 days to under 5 minutes**.
- Wrote **SQL** and **Snowpark Python** scripts to aggregate **Snowflake Cortex** usage data, helping IT management track internal AI usage.

Software Developer Fellow — September 2025 – Present
Longhorn Developers — *Austin, TX*

- Built **React/TypeScript** features for **Degree Audit Plus**, a Chrome extension with **50,000+ users**, using WXT and Tailwind CSS.
- Expanded the degree-audit parser from **3 to 6 formats**, adding a fallback parser and support for B.A. and B.S. programs.
- Prototyped course search using **IndexedDB**, filtering local course data by department, text, and course level.
- Worked with product managers to turn **Figma** designs into extension components and test new analytics features.

Software Developer — May 2025 – September 2025
CodeSprout — *Austin, TX*

- Integrated the **Piston API** with **Dockerized sandboxes** to execute user-submitted code in isolated environments.
- Designed **PostgreSQL schemas and indexes** for frequently executed application queries.
- Developed AI-assisted student features that received **80% positive teacher feedback** in beta demonstrations.

Projects

Aquarius Baskets | Live Demo

- Built a **FastAPI/Pandas portfolio backtester** for **2–25 assets** that models fractional shares, rebalancing, transaction costs, and drawdown against a benchmark.
- Added **3-fold walk-forward validation**, benchmarking optimized portfolios against **SPY** and equal-weight baselines on out-of-sample returns, win rate, and drawdown.
- Built a **LangGraph** portfolio copilot with **pgvector** retrieval and persisted chat history to generate allocation-constrained basket proposals.
- Deployed a **Next.js/FastAPI** app on **Google Cloud Run** with Supabase OAuth, **Redis** caching, and persistent portfolios.

Real-Time BTC Analytics Terminal | Link

- Built a real-time market analytics stack with a stateful **C++20** ingestion gateway, asynchronous **FastAPI** backend, and **Next.js** dashboard, updating **9 assets** every second.
- Unified **4 WebSocket/REST feeds** into one event stream while keeping the backend under **60 MB** of memory.
- Implemented a **C++** analytics engine for velocity, volatility, spreads, and cross-asset signals, with deterministic replay tests for state transitions.

Palladium Market Intelligence Agent | Link

- Built a **Python/PyTorch** pipeline analyzing palladium returns, volume, rolling volatility, and volatility regimes through z-scores and a trained market-regime classifier.
- Developed a **multi-task LSTM** on normalized 60-day OHLCV sequences to predict price direction, five-day realized volatility, and candle-anomaly probability with chronological validation and early stopping.
- Designed a parallel **LangGraph multi-agent workflow** where supply, demand, and macro agents analyze market news before a judge agent produces attributed research using the **OpenAI API**.

Dynamic Memory Allocator | *Course Project, C*

- Implemented a **C** memory allocator over **sbrk** with segregated free lists, **16-byte alignment**, block splitting, and coalescing, with constant-time free-list insertion and removal.

Technical Skills

Languages: Python, C++, TypeScript, JavaScript, SQL, Java
Frameworks / Data: FastAPI, Flask, React, Next.js, Pandas, PostgreSQL, Redis, Snowflake
Cloud / Tools: AWS, GCP, Docker, Terraform, Supabase, Git, Linux, LangGraph